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/* Name-Nirmayee kadam
/PRN-25070521134/
SECTION-C-I
Value Added Problem-2 */
#include <stdio.h>
#include <string.h>

#define MAX 1000000

char stack[MAX];
int top = -1;

// Push function
void push(char c) {
    stack[++top] = c;
}

// Pop function
char pop() {
    if (top == -1)
        return '\0';
    return stack[top--];
}

// Check if matching pair
int isMatchingPair(char open, char close) {
    if (open == '(' && close == ')') return 1;
    if (open == '{' && close == '}') return 1;
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    if (open == '[' && close == ']') return 1;
    return 0;
}

int isBalanced(char *s) {
    for (int i = 0; s[i] != '\0'; i++) {

        // If opening bracket, push
        if (s[i] == '(' || s[i] == '{' || s[i] == '[') {
            push(s[i]);
        }

        // If closing bracket
        else if (s[i] == ')' || s[i] == '}' || s[i] == ']') {

            if (top == -1) // No matching opening
                return 0;

            char topElement = pop();

            if (!isMatchingPair(topElement, s[i]))
                return 0;
        }
    }

    // If stack is empty → balanced
    return (top == -1);
}

```

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int main() {

    printf("Name :Nirmayee kadam \n");

    printf("PRN :25070521134\n");

    printf("Section :C-1\n");

    printf("Value Added Problem : 2\n");

    printf("-----\n");

    char s[MAX];

    printf("Enter expression: ");

    scanf("%s", s);

    if (isBalanced(s))

        printf("true\n");

    else

        printf("false\n");

    return 0;

}

```

main.c	Run	Output
<pre> 1 /* Name-Nirmayee kadam 2 /PRN-25070521134/ 3 SECTION-C-I 4 Value Added Problem-2 */ 5 #include <stdio.h> 6 #include <string.h> 7 8 #define MAX 1000000 9 10 char stack[MAX]; </pre>		<pre> ^ Name :Nirmayee kadam PRN :25070521134 Section :C-1 Value Added Problem : 2 ----- Enter expression: ({[]}) true === Code Execution Successful === </pre>